University of Augsburg

Anwendungen der Data Science (SS 2026)

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Exercise Sheet: Heart Disease Across Populations

Background

Cardiovascular diseases are among the leading causes of mortality worldwide. Clinical data are often collected from different regions, where patient populations, medical practices, and environmental factors may vary.

In this exercise, you are given a dataset that combines patient-level observations from four regions: *Hungary, Cleveland, Switzerland, and VA Medical Center*. Each observation contains clinical measurements and an indicator of whether the patient has heart disease.

The objective is to investigate whether these data can be treated as coming from a single homogeneous population, and to construct a model for assessing patient risk.

The dataset used in this exercise is provided as CSV file: `heart.csv`.

Variables

- **age**: Age of the patient (years)
- **sex**: Sex (0 = female, 1 = male)
- **cp**: Chest pain type
- **trestbps**: Resting blood pressure (mm Hg)
- **chol**: Serum cholesterol (mg/dl)
- **fbs**: Fasting blood sugar (> 120 mg/dl)
- **restecg**: Resting electrocardiographic results
- **thalach**: Maximum heart rate achieved
- **exang**: Exercise-induced angina
- **oldpeak**: ST depression induced by exercise
- **slope**: Slope of peak exercise ST segment
- **ca**: Number of major vessels

- **thal**: Thalassemia
- **clinical_index**: A summary score combining several clinical measurements
- **target**: Indicator of heart disease (1 = presence, 0 = absence)
- **country**: Region of origin

Tasks

Task 1: Understanding the data

(a) Inspect whether all variables are recorded consistently across regions.

(b) Identify variables that contain missing values and describe how they appear in the dataset.

Hint: Before comparing populations, ensure that the data is comparable.

Task 2: Comparing populations

(a) Consider variables such as age, cholesterol, and maximum heart rate. Do these appear similar across regions?

(b) Is the prevalence of heart disease comparable across the four regions?

(c) Select two clinically relevant variables and examine whether their relationship with heart disease appears consistent across regions.

(d) One of the variables represents a summary of several clinical measurements. Discuss how including such a variable alongside its components may affect the interpretation of coefficients.

Task 3: Quantifying risk

A physician would like to estimate the probability that a patient has heart disease based on clinical measurements.

(a) Propose a model that maps patient characteristics to a value between 0 and 1.

(b) Explain how the parameters of your model should be interpreted.

(c) Identify which variables increase and which variables decrease the risk.

(d) Compute the estimated probability for a patient with given characteristics.

Hint: The model should reflect the probabilistic nature of the outcome.

Task 4: Stability across regions

(a) Investigate whether combining regions improves prediction.

(b) Are the effects of key variables similar across regions?

(c) Which variables behave inconsistently?

(d) Discuss whether a single global model is appropriate.

Task 5: Unknown structure in the data

Suppose that the region labels were not available.

(a) Can you group patients based solely on their clinical characteristics?

(b) Compare these groups with the known regions.

(c) Which grouping appears more informative:

- region-based grouping, or
- data-driven grouping?

Hint: Not all structure in the data is explicitly labeled.

Task 6: Interpreting differences

(a) Suppose one region has higher cholesterol levels but lower disease prevalence. How would you interpret this?

(b) Can differences across regions always be attributed to biological factors?

(c) Provide at least two alternative explanations.